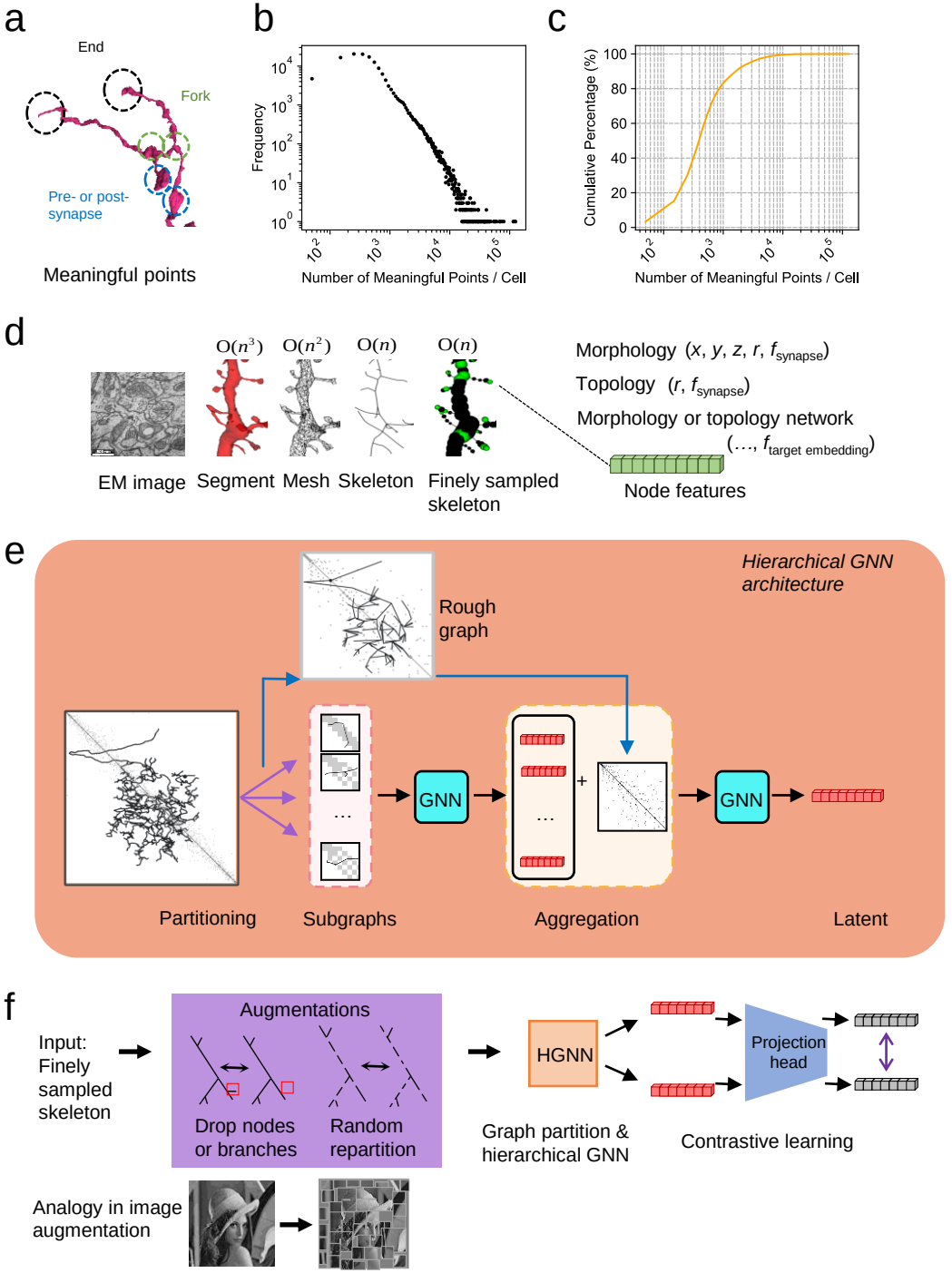
a
End
Fork
Pre- or post-synapse
Meaningful points
b
Frequency
Number of Meaningful Points / Cell
c
Cumulative Percentage (%)
Number of Meaningful Points / Cell
d
O(n³)
O(n²)
O(n)
O(n)
Morphology (x, y, z, r, f_synapse)
Topology (r, f_synapse)
Morphology or topology network
(…, f_target embedding)
EM image
Segment
Mesh
Skeleton
Finely sampled skeleton
Node features
e
Hierarchical GNN architecture
Rough graph
GNN
GNN
Partitioning
Subgraphs
Aggregation
Latent
f
Input: Finely sampled skeleton
Augmentations
Drop nodes or branches
Random repartition
HGNN
Graph partition & hierarchical GNN
Projection head
Contrastive learning
Analogy in image augmentation